# Foundations of Operations Research

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### Students

- Computer Science and Engineering: from **G** to **P** included
- Biomedical Eng., High Performance Computing Eng., Music and Acoustic Eng., Telecommunication Eng.: from **E** to **O** included

### Course mode

The lectures, the exercises and the computer labs will be held in the classroom, with no streaming.

### Course material

All information and material (lecture slides, exercises and computer labs) will be made available on WeBeep.

Slides designed to support the lectures, not to be sufficient for independent study.

## Schedule

- Tuesday 10.15-12.15: **lecture** (room T.1.1)
- Wednesday 10.15-12.15: **45' lecture** and **45' exercises** (room T.2.2)
- Monday 16.15-18.15: **computer labs**, dates TBA (room TBA)

## Instructors

- Lectures and exercises: Edoardo Amaldi (`edoardo.amaldi@polimi.it`)
- Computer labs: Maximiliano Cubillos (`maximiliano.cubillos@polimi.it`)

## Office hours

- At the end of the class or by appointment

### Exercises and computer labs

- Weekly sets of problems (assigned before hand) to be corrected in class the following week – interactive teaching.
- Computer lab sessions: 3 sessions of 70 minutes each and one assignment (dates TBA).

### Independent study

A couple of topics (e.g. LP sensitivity analysis) will be assigned for independent study prior to exercises or computer laboratory activities.

### Optional computational challenge

Tackle a simplified real-world problem with variants/extensions of the models and algorithms covered in the course.

Focus on some of the fundamental optimization models and methods to tackle complex decision-making problems.

**Syllabus**

1. Introduction
2. Graph and network optimization
3. Linear Programming (LP)
4. Integer Linear Programming (ILP)

Link to the detailed program

Time permitting: introduction to heuristic algorithms (greedy and local search).

## Course objectives

- Learn how to analyze a decision-making problem, build an optimization model, identify an appropriate algorithm to solve it and interpret the output.
- Understand some of the main optimization methods (e.g. for graph optimization, project planning, LP and ILP).
- Learn how to use a modeling language (Python) and a state-of-the-art optimization solver (CPLEX or Gurobi).

**Prerequisites:**

Basic concepts from Calculus, Linear Algebra and Algorithms.

## Evaluation

- A written exam covering all the material presented in the lectures, exercise/computer lab sessions and assignments (worth **31 points**).

  Problems aimed at evaluating the level of (applying) knowledge and understanding of the concepts, models, results and methods covered.

  Old exam questions will be available.

- Three short assessments (quizzes) will be carried out during the semester (worth 2 points), dates TBA.

  The students who completed the quizzes can **skip a question** (worth 2 points) in the written exam.

## Teaching material

- Slides of the lectures and material for the exercise/computer lab sessions on `WeBeep`.
- For a similar/complementary treatment:
  - M. Fischetti, *Introduction to Mathematical Optimization*, 2019
  - F. Hillier, G.J. Lieberman, *Introduction to Operations Research*, McGraw-Hill, 2010

  Graph algorithms:
  - R.K. Ahuja, T.L. Magnanti, J.B. Orlin, *Network flows. Theory, algorithms and applications*, Prentice Hall, 1993

  Linear Programming and introduction to Integer Programming:
  - D. Bertsimas, J. Tsitsiklis, *Introduction to Linear Optimization*, Athena Scientific, 1997